1. Osteoporosis: overview

NICE CG146, 2017; NICE TA160 and TA161, 2018; TA464, 2019; TA204, 2010; TA791, 2022; SIGN 142 (2021); NOGG 2021

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The 3 guidelines on osteoporosis (NICE, SIGN and NOGG) all say different things! This is in part because evidence is lacking. NICE also looks to economic modelling. Whichever you follow, avoid switching between them halfway.

Basic principles for all guidance: Identify who is at risk → Quantify risks using scoring tool (FRAX/QFracture) unless high risk. Do DXA if indicated → Treat if appropriate: drug choice according to guideline chosen

Identify who is at risk

All the major guidelines suggest a case-finding approach to identify those at increased risk of osteoporosis. However, the 3 major guidelines (NICE, NOGG and SIGN) all suggest slightly different populations to look for osteoporosis in! **What about population screening? No!** Insufficient evidence (National Screening Committee, 2019).

How do we identify those at risk?
There are many risk factors for osteoporosis. The FRAX tool is a good prompt for key risk factors, and these risks are included in all the major guidelines.

See box for risk factors assessed in the FRAX tool: →

Risk factors assessed in FRAX tool:
- Age, sex, BMI
- Smoking, alcohol (≥3U/d)
- Previous fracture
- Parent with hip fracture
- Rheumatoid arthritis
- Steroids (any dose, by mouth, ≥3m)
- Secondary osteoporosis (including premature menopause (<45y), T1 diabetes, chronic liver disease and chronic malabsorption/malnutrition)

Which risk factors do the 3 guidelines use?
NICE risk factors roughly map to the FRAX list.
SIGN adds a few more.
NOGG has a massive list!

Quantify risk

How do we risk assess those identified as being at increased risk?

Most people → **Use fracture risk assessment tool (FRAX or QFracture) to assess risk** → **Do DXA/treat as directed by score**

A few proceed straight to DXA → **Varies by guideline, but tends to be high-risk groups:**
- **SIGN: had fragility fracture**
- **NICE: <40y with major risk factors:**
  - Fragility fracture
  - Glucocorticoids (prednisolone >7.5mg/d for ≥3m)
  - Untreated premature menopause

There are 2 scoring tools (FRAX/QFracture): they give a percentage risk of fracture in next 10y.

Which to use?
- NICE uses FRAX – it says use either, but FRAX needed for who/when to treat.
- NOGG prefers FRAX as it links to NOGG treatment guidance.
- SIGN prefer QFracture (due to comprehensive list of risk factors).

At Red Whale, we think FRAX is easiest because later in the process you can use the FRAX site to guide what to do with the answer!

The scoring tools give you 2 results:
- Risk of hip fracture.
- Risk of major osteoporotic fracture.

Which fracture risk do we use?
- NOGG says use highest score to decide treatment.
- NICE says use major osteoporotic fracture risk.

When to treat

- **If you chose to follow NICE or NOGG,** treatment is based on FRAX score, which is mapped onto NOGG intervention thresholds (namely lifestyle; use DXA to decide if to treat; treat without DXA) (see overleaf).
- **If you chose to follow SIGN,** treatment is based on:
  - Primary prevention: do QFracture score. If ≥10%: do a DXA and treat on basis of result.
  - Those with a fracture (secondary prevention): treatment based on DXA results.

Which drug to use

- Oral bisphosphonates are first line in most guidance.
- Check renal function, vitamin D and calcium, and arrange dental check, before starting.
- IV bisphosphonates or denosumab should be considered if oral treatment not tolerated.
- In hip fracture trials, IV bisphosphonates reduced mortality at 3y and reduced fracture recurrence. For severe vertebral osteoporosis, consider referral for newer treatments.

2. FRAX/NOGG in action

Screen shots from the FRAX website – link below

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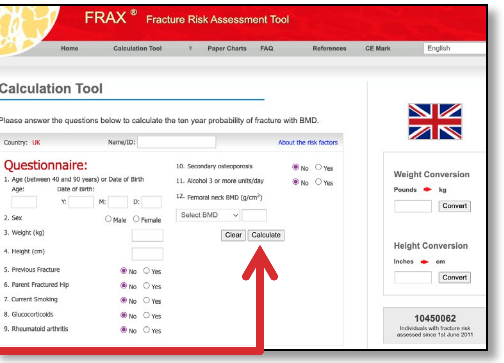
Using FRAX/NOGG graphs to risk assess/manage those with/at risk of osteoporosis

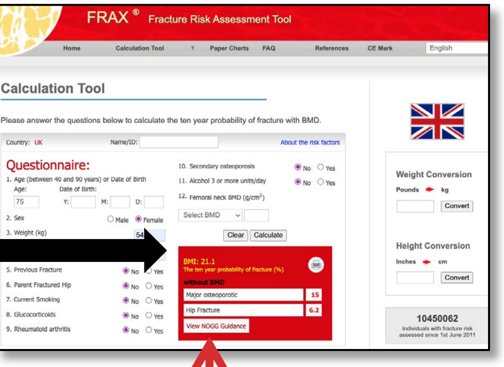
We are not expecting you to be able to read all the words on the screenshots on this page – the images are there so that first-time users can follow along and are reassured they are on the right page!

- **Go to the FRAX website**
  - **For those born in the UK, use the UK tool** (UK flag on right hand side of page).
  - **If born outside UK, find the page for the country of birth**.



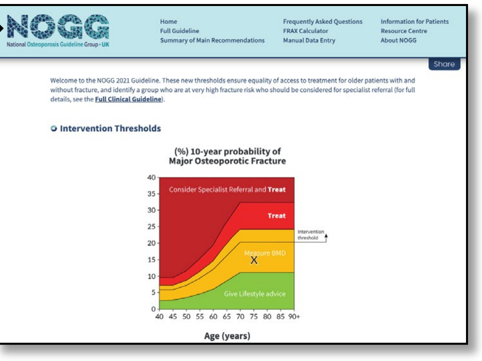
- **Enter data on FRAX** – at this stage, you usually won't have a bone density scan so leave that blank.
- **If they have T2 diabetes**, tick the rheumatoid arthritis box.
- **Click 'Calculate'**

- A red box will appear, giving the risk (over 10y) of major osteoporotic fracture and of hip fracture.



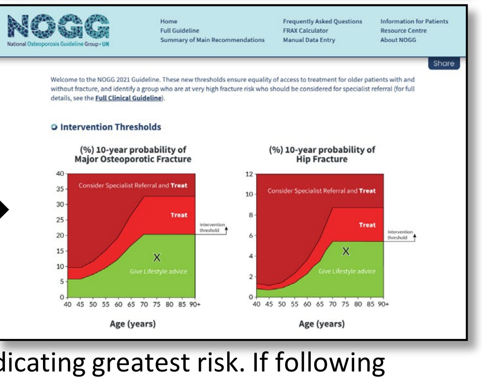
- **Click the 'View NOGG guidance' button within the red box** (only appears if you are on UK page).

This takes you through to the NOGG site, and a graph will be shown.



- **Look for the cross, and follow the instructions for the area in which the cross falls:**
  - Green: lifestyle alone.
  - Yellow: do DXA.
  - Red: treat.
  - Deep red: treat, consider specialist referral.
- **You will also see a line marked 'intervention threshold':** NOGG recommends starting treatment immediately (and requesting DXA) for this group.

- **If a DXA is indicated, when the result comes back, re-enter the data on FRAX, entering the BMD result** (for most, this is likely to be the T-score).
- **Click the 'View NOGG guidance' button.** Now, you will get TWO graphs, each with red and green on them.
- **Look again for the cross and manage as indicated:**
  - Green: lifestyle alone.
  - Red: treat.
  - Deep red: treat, consider specialist referral.



What if crosses are on different colours? If following NOGG, use cross indicating greatest risk. If following NICE, always look at risk of major osteoporotic fracture (left hand graph).

3. Investigations, lifestyle, steroids

NICE CG146, 2017; NICE TA160 and TA161, 2018; TA464, 2019; TA204, 2010; TA791, 2022; SIGN 142 (2021); NOGG 2021

Looking for underlying causes

NOGG suggests: **FBC, creatinine, LFT, bone profile, TFT, CRP or ESR, vitamin D, phosphate.**
NOGG also advises spine X-rays in confirmed osteoporosis to look for asymptomatic vertebral fractures as this might affect treatment choice. *See main article for full details.*
Consider investigating for underlying secondary causes such as endocrine disorders, malabsorption and metabolic bone disorders in certain groups (*NOGG does not state who*). Our list is based on the Oxfordshire and Nottinghamshire osteoporosis guidelines.

In those with fragility fracture/osteoporosis, look for secondary underlying causes if:

- Man of any age.
- Woman pre-menopause or in the early stages of menopause
- Fragility fractures at unusual sites or vertebral fractures painful >6w or worsening.

A list of suggested investigations for these groups is included in the main article.

Lifestyle advice

Calcium and vitamin D (based on SIGN and NOGG)

- Assess vitamin D and calcium status.
- **Try to meet daily calcium requirements through dietary means, aiming for 700mg/d.** Calcium calculators help do this: e.g.: CGEM Calcium Calculator (ed.ac.uk)
- **Consider calcium supplementation if dietary intake is inadequate and cannot be increased.** Do not take these within 2h of bisphosphonates (reduces effectiveness).
- **Consider offering vitamin D supplements daily if vitamin D deficiency or risk factors for deficiency are present** (NOGG recommends 800IU daily in these groups).

Other measures:

- Stop smoking and reduce alcohol to <2u/day (NOGG).
- Weight-bearing activity may slow the decline of bone density.
- Balance training, flexibility and stretching exercises may reduce the risk of falls.
- Do a falls risk assessment (those at risk should be referred to a falls exercise programme).
- Aim for a balanced diet to maintain BMI 20–25. No particular diet is better than another.
- **Do NOT** recommend phyto-oestrogens/vitamin K supplements/acid-balancing diets (SIGN).

Duration of therapy

Depends on which guideline you follow (see relevant GEMS page), but, as rule of thumb:
Alendronate: usually 5y, longer if higher risk (e.g. older, previous hip/vertebral fractures).
Zoledronate IV: usually 3y (sometimes longer if high risk).
After period of treatment, review (patient, risks, +/- DXA). Drug holidays (treat, pause, re-start treatment) are not widely recommended. **Do NOT stop denosumab without specialist review.**

Steroids and bone risk

Alendronate, risedronate, zoledronate denosumab and teriparatide are all now licensed for bone protection in those using steroids.

In anyone (not just those with osteopenia/osteoporosis) starting on steroids, check which group your patient falls into:

High-risk groups

- Anyone with previous fragility fracture.
- Any woman ≥70y on ANY glucocorticoid dose.
- Postmenopausal women, or men ≥50y:
  - On high doses (equivalent to ≥7.5mg/d prednisolone over 3m).
  - If FRAX score is in the yellow section but above the intervention line.

- **Calculate FRAX.**
- **Click through to NOGG to see if DXA indicated.**
- **Start bone protection without waiting for DXA.**

Everyone else
*(including those on **intermittent steroids** if they have taken ≥30mg/day for ≥4w within any 3m period).*

- **Calculate FRAX and follow resulting NOGG guidance.**
- **Treat according to DXA.**

If steroids stopped, reassess FRAX to decide if bone treatment can also be stopped.

4. NOGG on osteoporosis

NOGG 2021

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When should we look for it?

Risk assess anyone with a clinical risk factor.

NOGG recommends a case-finding approach (actively seeking out) in all those with risk factors – the list is extensive and may be unrealistic considering pressures in primary care!

- Postmenopausal women
- Men ≥50y
- Low BMI <18.5
- Previous fragility fracture
- Parental hip fracture
- Smoking
- Alcohol ≥3U/day
- HIV
- Rheumatoid arthritis
- Secondary osteoporosis
- Height loss ≥4cm
- Thoracic kyphosis

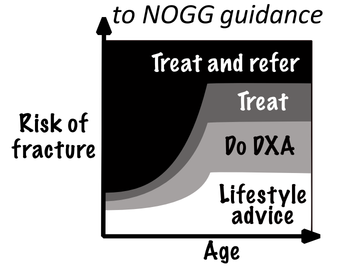
- Glucocorticoids
- Inflammatory diseases (inflammatory arthritis, connective tissue disorders)
- Endocrine diseases (diabetes, hyperthyroidism, Cushing's syndrome, premature ovarian insufficiency, hypogonadism)
- Haematological disorders/malignancy (myeloma, thalassaemia)
- Muscle disease (muscular dystrophy, sarcopenia)
- Lung disease (asthma, COPD, cystic fibrosis)
- Neurological disease (PD, epilepsy, MS, stroke, depression, dementia)
- Nutritional: nutritional deficiency (calcium and vitamin D deficiency), bariatric surgery, risk of malabsorption (IBD, coeliac disease)
- Medications: immunosuppressants, glitazones, antiepileptics (phenytoin, carbamazepine), gonadal hormone suppressants

How to quantify risk

Use the FRAX scoring tool. Calculate FRAX then click 'view NOGG guidance' to see result:

FRAX tool will determine whether to offer lifestyle advice, do DXA (then reassess) or treat. If DXA recommended, once results of this back, recalculate FRAX, adding in DXA result for femoral neck (usually the T-score). (If lumbar spine BMD lower than femoral neck, increase probability of major osteoporotic fracture by one tenth for each T-score of difference, e.g. Lumbar spine T-score -2.5, hip -1.5, increase score by a tenth. Do not revise down if BMD scores are the other way around).

Example of FRAX mapped to NOGG guidance



When using FRAX:

- **Use the tool for the COUNTRY OF BIRTH** (even if lived in UK most of life).
- **If type 2 diabetes:** tick 'rheumatoid arthritis box' on FRAX.
- **Recurrent falls (≥2 in past 12m)**: increase risk of fracture by one-third.
- **If on >7.5mg glucocorticoids daily:** increase probability of fracture by 15% for major osteoporotic fracture and by one-fifth for hip fracture (even after ticking the 'steroids' box).

When and how to treat

TREAT IF: 'NOGG graph' indicates treatment needed.

TREATMENT OPTIONS:

- **For hip fracture, use IV zoledronate** (reduces mortality at 3y and fracture recurrence).
- **For other fractures, use oral bisphosphonates or IV zoledronate.**
- **For those at very high risk, NOGG suggests treatment and referral.** May be considered for anabolic drugs (teriparatide, romosozumab) (especially if history of vertebral fracture).

DURATION OF TREATMENT:

- >70y
- Previous hip/vertebral fracture
- ≥2 vertebral fractures
- Current glucocorticoid use (prednisolone ≥7.5mg/d ≥3m or equivalent)

ANY of these → Counsel that duration will be for: **oral therapy: 10y**, **IV therapy: 6y** (or if indication is glucocorticoids, until steroids stopped)
Once course complete, therapy usually stopped (lack of evidence): individual patient discussion.

NONE of these → Counsel that duration will be for: **oral therapy: 5y**, **IV therapy: 3y**
Once treatment course complete, re-do FRAX and BMD

- **If below NOGG treatment threshold:** consider pausing drug and review again after 1.5y (ibandronate and risedronate) or 3y (alendronate, zoledronate), or sooner if new fracture.
- **If above NOGG treatment threshold:** continue treatment for further 5y for oral and 3y for IV. Review at the end of 10y/6y treatment: most will stop at this stage: individual patient discussion

Do NOT stop denosumab without specialist review (MHRA 2020)

5. NICE on osteoporosis

NICE CG146, 2017; NICE TA160 and TA161, 2018; TA464, 2019; TA204, 2010; TA791, 2022

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We found it fiendishly difficult to represent NICE's patchwork guidance on this page! NICE guideline on osteoporosis (CG146) only covers risk assessment (!). We then had to turn to the NICE quality standards and technology appraisals to work out WHEN and WHO to treat, and the indications for each drug.

When should we look for it?

Risk assess the following groups:

All women >65y
All men >75y

Women aged 50–65y **Men aged 50–75y**
Risk assess if they have:

- Low BMI <18.5.
- History of falls or fragility fracture.
- Parental hip fracture.
- Smoking.
- Alcohol: women ≥14U/w, men ≥21U/w.
- Rheumatoid arthritis.
- Glucocorticoids (prednisolone >7.5mg/day. for ≥3m or equivalent).
- Secondary osteoporosis.

These are the FRAX risk factors, but FRAX uses a different cut-off for alcohol and steroids

<50y
Do NOT routinely risk assess

Special group: <40y with major risk factors

- Glucocorticoids (prednisolone >7.5mg/day for ≥3m or equivalent).
- Fragility fracture.
- Untreated premature ovarian insufficiency (premature menopause).

CONSIDER DXA rather than using scoring tool.

How to quantify risk

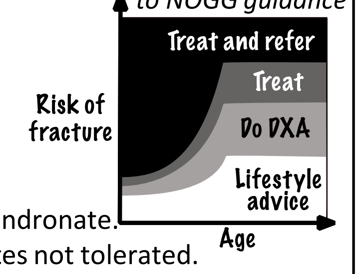
Use a scoring tool (unless in special group, outlined above): NICE says either FRAX or QFracture can be used, but NICE treatment thresholds are based on FRAX so use this!
Consider DXA if score is close to treatment thresholds or in special group, outlined above.
Reassess fracture risk only after interval of 2y or if risks change.

When and how to treat

TREAT IF:

Calculate FRAX, then click 'view NOGG guidance' to see result:
FRAX tool will determine whether to offer lifestyle advice, do DXA (then reassess) or treat. If DXA recommended, once results of this back, recalculate FRAX, adding in DXA result for femoral neck (usually the T-score).

Example of FRAX mapped to NOGG guidance



TREATMENT OPTIONS:

- Oral bisphosphonates for most: alendronate, risedronate, ibandronate.
- IV bisphosphonates for very high risk or if oral bisphosphonates not tolerated.
- Teriparatide only in special situations (usually severe disease).

(When NICE wrote its denosumab guidance, the drug was expensive and with less evidence for use; if the guidance was written now, it might play a more prominent role.)

DURATION OF TREATMENT:

After 3y zoledronate or 5y alendronate, ibandronate or risedronate, review fracture risk:

- **For high-risk patients** (age ≥75y, previous hip or vertebral fracture, fracture while on bone-modifying treatment or currently on glucocorticoids ≥7.5mg/d), continue treatment (NICE does not say how long for!) (DXA not needed). Take into account drug used, patient choice and life expectancy. NICE does not advise drug holidays.
- **For everyone else,** consider repeat DXA and base treatment decision on T-score.

Do NOT stop denosumab without specialist review (MHRA 2020)

6. SIGN on osteoporosis

SIGN 142 (2021)

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SIGN guidance on osteoporosis includes a complex flowchart for treatment choice. We have included this in full at the end of the main article.

When should we look for it?

Risk assess the following groups:

Anyone with a major clinical risk factor

Known low bone-mineral density
Oral steroid use
HIV*
Untreated premature menopause

>50y with a clinical risk factors

Parental osteoporosis
Alcohol >3.5U/day
Low BMI (<20)
Smokers
Rheumatoid arthritis/SLE
Inflammatory bowel disease
Moderate/severe CKD
Neurological diseases (MS, Parkinson's, Alzheimer's, CVA)
Epilepsy if living in an institution
Medication: SSRIs, PPIs, glitazones, enzyme-inducing antiepileptics, hormone blockers
Asthma
Diabetes
Endocrine disease
HIV*
Chronic liver disease

**When we asked SIGN about HIV, it said it was likely this was only really a risk >50y, and this was a mistake – it should probably only appear in the >50y group!*

How to quantify risk

Use the QFracture scoring tool (preferred because more comprehensive list of risk factors).

When and how to treat

TREAT IF:

- **Primary prevention (never had a fragility fracture):**
  - If QFracture risk ≥10%, arrange DXA scan and follow flowchart at the end of main article.
  - If QFracture risk <10%, offer lifestyle advice.
- **Secondary prevention (had a fragility fracture):**
  - Arrange DXA scan.
  - Start treatment (following flowchart in main document) while waiting for results.

TREATMENT OPTIONS:

Treatment depends on T-score and site of any fragility fracture. See flowchart in main article.

DURATION OF TREATMENT:

For those on bisphosphonate/denosumab: check DXA response at 3y.

Duration of treatment depends on treatment recommended and indication! Probably best to follow flowchart at the end of the main article, but, in brief:

Alendronate: 5y for most. 10y in postmenopausal women, especially if high risk of vertebral fracture.

Risedronate: up to 7y in postmenopausal women.

Zoledronate: used in 2 ways by SIGN:

1. Annual infusions for 3y, then drug holiday and review at year 5 (post hip fracture).
2. 18-monthly infusions for 6y, then review (for osteopenia).

Do NOT stop denosumab without specialist review (MHRA 2020)